

Recursive Object Serialization Engine

In distributed systems and RESTful API development, converting complex Python object graphs into structured dictionary representations is a critical requirement. A generic serialization engine allows developers to automate this transformation using decorators, handling nested objects and collections while maintaining precise control over attribute visibility.

Implement a class decorator `serializable` that dynamically injects a `to_dict()` method into any decorated class. The engine must support recursive serialization: if an attribute is another serializable object (or a list of such objects), it must automatically invoke their `to_dict()` methods to produce a fully nested dictionary output.

Example 1 Input:

```
SERIALIZE User ["id", "name"]
```

Output:

```
{"id": 1, "name": "Alice"}
```

Explanation: The `User` class is decorated to only include `id` and `name` in its dictionary output. The `email` field is excluded as it was not specified in the `fields` list.

Example 2 Input:

```
SERIALIZE Group ["id", "users"]
```

Output:

```
{"id": 101, "users": [{"id": 1, "name": "Alice"}, {"id": 2, "name": "Bob"}]}
```

Explanation: The `Group` object contains a list of `User` objects. The recursive engine identifies that the items in the `users` list are themselves serializable and converts them into dictionaries automatically.

Function Parameter:

- `fields` (list of str): The attribute names to include in the serialization.

Returns

- dict: The structured dictionary representation of the object.

Constraints

- The original class identity must be preserved (modify the class, don't return a new one).
- Handle both single nested objects and lists of objects recursively.
- Assume all specified fields exist on the instance during serialization.

Input Format for Custom Testing Input from stdin contains: `SERIALIZE <class_name>`
`<fields_json>`.

Sample Case 0 Sample Input

```
SERIALIZE User ["id"]
```

Sample Output

```
{"id": 1}
```

Explanation A minimal serialization including only the primary identifier of the User object.